

Balireddy Mahalakshmi

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Professional Summary

AI & Machine Learning graduate with hands-on experience in designing, developing, and deploying machine learning and deep learning models to solve real-world problems. Proficient in Python, SQL, TensorFlow, Keras, OpenCV, and data preprocessing, with strong knowledge of supervised learning, NLP, predictive analytics, and model evaluation techniques. Experienced in building scalable ML pipelines, automating data workflows, and deploying cloud-based solutions using AWS. Skilled in model optimization, data visualization, and collaborative development using Git/GitHub. Strong analytical, problem-solving, and communication abilities with a focus on delivering efficient, scalable, and production-ready AI solutions.

Technical Skills

- **Programming Languages:** Python, SQL (MySQL), C, Data Structures & Algorithms (DSA).
- **Machine Learning:** Model Development, Deep Learning Architectures, NLP, Data cleaning & preprocessing.
- **Libraries & Frameworks:** TensorFlow, Keras, OpenCV, Pandas, NumPy, Matplotlib.
- **Tools & Cloud:** AWS (Certified), FastAPI, Postman, Git, GitHub, Power BI, VS Code, MS Excel.
- **Soft Skills:** Collaborated with cross functional teams, Documented ML Workflows, communicated findings using data visualization tools

Professional Experience

Machine Learning Intern | FabHost Tech Solutions | Jan 2024 – Mar 2024

- Developed and deployed supervised ML models (Random Forest, XGBoost) using Scikit-learn, performing hyperparameter tuning with GridSearchCV and evaluating performance using Precision, Recall, F1-score.
- Automated **data-cleaning pipelines** by developing custom Python scripts, reducing manual data preparation time by 20% and ensuring high data integrity.
- Streamlined ML workflows by implementing modular code and version control, improving the scalability and reproducibility of team-wide experiments.

Projects

Netra: AI-Based Real-Time Assistive Vision System

- Developed a **multi-modal AI pipeline** integrating **YOLOv8** for object detection, **EasyOCR** for text recognition, and **MobileNetV2** for high-accuracy currency classification.
- Architected a **Client-Server infrastructure** using **Android (CameraX)** and **FastAPI**, enabling real-time frame capture and high-speed model inference via a modular backend.
- Containerized the **FastAPI backend** using **Docker** and deployed on **AWS EC2** for scalable real-time inference, handling concurrent API requests with **sub-200ms latency**.
- Reduced inference latency by 35% through OpenCV preprocessing and **asynchronous API (Retrofit)** handling to deliver seamless, low-latency audio feedback via Text-to-Speech (TTS).

IoT-Based Weather Forecasting System

- Architected a **real-time data pipeline** using **ESP32** sensors and **TensorFlow**, achieved 91% forecasting accuracy on 12-month time-series data using LSTM.
- Engineered an **interactive dashboard** by integrating **Google Maps API**, enabling real-time location-based data visualization and sensor mapping.
- Developed **automated alerting workflows** using **IFTTT protocols** and asynchronous data streams to provide instant mobile notifications based on environmental thresholds.

Education

Bachelor of Technology (B.Tech) in AI & Machine Learning

Saveetha Engineering College | Chennai, TN | 2021 – 2025 | **CGPA: 8.0/10**

Certifications

- AWS Certified Cloud Practitioner | Amazon Web Services (EC2, S3, IAM, Cloudwatch)
- Machine Learning with Python | IBM / Udemy
- Python Programming Certification | FabHost Tech Solutions